

Learnable Experts Orchestration

With Multi Armed Bandits

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Специализация: Интеллектуальный анализ данных

Model orchestration as Multi Armed Bandit

Problem:

Modern ML systems require supporting multiple learnable models operating in an online regime. At each decision step, this entails (i) selecting a model for prediction and (ii) selecting models for training. A theoretical framework for this problem needs to be developed.

Objective:

To formalize the orchestration problem for learnable models and propose an algorithm with theoretical guarantees.

Solution:

- 1) Formalize the problem as a multi-armed bandit (MAB) problem and define regret as the performance metric.
- 2) Leverage theoretical guarantees from optimization methods and concentration inequalities to derive confidence intervals for the algorithms.
- 3) Employ the confidence intervals within a UCB-type algorithm.

Multi Armed Bandits

Setup

- Given K arms/actions $\{a_i\}_{i=1}^K$
- Space of rewards (e.g. $[0,1]$)
- Reward functions: arm a pull $\rightarrow r_a^t$ (e.g. random variable)
- Decision time horizon T
- Per round training budget M

Protocol

- For $t = \overline{1, \dots, T}$:
- Choose arm a_t and pull.
- Receive $r^t := r_{a_t}^t$.

Aim

$$R(T) = T \max_{a \in K} \mathbb{E} r_a - \sum_{t=1}^T \mathbb{E} r_t \rightarrow \min$$

Best fixed arm

Accumulated
reward

UCB algorithm

Upper Confidence Bound

- Build indexes $\forall k \in [K] : \text{LCB}(k, t, \delta) \leq \mathbb{E}r_k \leq \text{UCB}(k, t, \delta)$

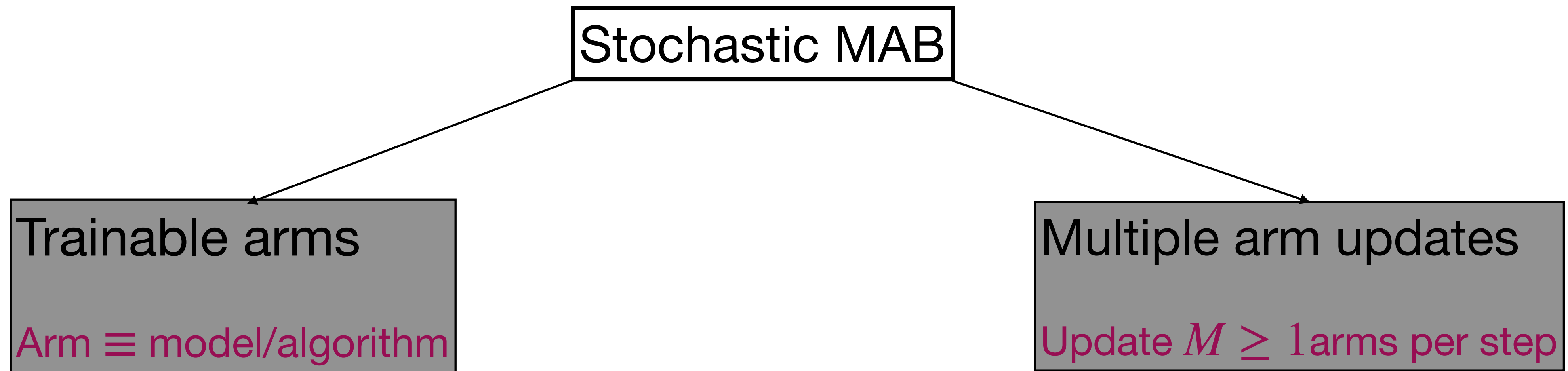
- $$\text{UCB}(k, \tau, \delta) = \frac{1}{\tau} \sum_{i=1}^{\tau} r_k^i + c \sqrt{\frac{\log \frac{1}{\delta}}{\tau}}$$

Algorithm

- For $t = \overline{1, \dots, T}$:
 - Choose arm $a_t = \arg \max_{k \in [K]} \text{UCB}(k, \tau_k, \delta)$
 - Receive $r^t := r_{a_t}^t$.

Multi Armed Bandits

Extension



Arm examples

- Model, trained with online optimization algorithm, (e.g. OGD, Cat boost).
- Arm itself Multi armed bandit.

Example

Recommendation system

Покупают вместе

КОВРИК
для мыши

что ладно, я уважаю твоё
неправильное мнение

220x180x3 мм

408 Р

Коврик для мышки приколом
мем Котик Штош ладно я...

★ 4.9 ● 3 080 отзывов

12 декабря

Наклейка

на карту

Распродажа

150 Р 450 Р -66%

1046 шт осталось

Наклейка на карту
Взрывной кот

★ 4.8 ● 6 008 отзывов

Послезавтра

MOFU SAND
НАКЛЕЙКИ

Распродажа

91 Р 699 Р -86%

3299 шт осталось

Наклейки Котики Mofu
Sand 50шт комплект 1

★ 5.0 ● 10 576 отзывов

Завтра

КОВРИК
для мыши

не то чтобы понял,
но понял

220x180x3 мм

408 Р

Коврик для мышки с
приколом мем котик Не то...

★ 5.0 ● 1 479 отзывов

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Soon 3

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TAKARA TOMY 20cm

Main recommendation

Others

Notations & Assumptions

Notations

- Data: $\xi_t \sim D$,
- Arm: $w_k \in \mathbb{R}^n$, At step t for arm k k-th model prediction
- Loss function: $\ell_k^t(w) = \ell(g_k(w, \xi_t), \xi_t)$.

Assumption 1: i.i.d. data $\xi_t \sim D$,

Assumption 2: $\forall k \in [K] : \forall \tau \geq 1 :$ Running loss $\sum_{i=1}^{\tau} \ell_k^i(w^i) - \min_w \sum_{i=1}^{\tau} \ell_k^i(w)$ Empirical loss minimizer $\leq U_k(\tau)$.

Trained on τ datapoints

Notations & Assumptions

Protocol

- For $t = \overline{1, \dots, T}$:
 - Choose arms set $S_t = \{a_j\}; |S_t| = M$,
 - Choose decision arm $a_t \in S_t$,
 - Receive data ξ_t , suffer loss $\ell_{a_t}^t(w_{a_t}^t)$,
 - Observe losses for $a \in S_t$.

Note: Actually, in algorithm we use some "smoothing" trick

Aim

$$R(T) := \sum_{t=1}^T \ell_{a_t}^t(w_{a_t}^t) - T \min_{k \in [K]} \min_{w_k} \mathbb{E}_{\xi} \ell(g_k(w, \xi), \xi) \rightarrow \min.$$

Best fitted model

Expected loss

M-LCB algorithm

Confidence bounds

Assumption 2: $\forall k \in [K] : \forall \tau \geq 1 : \sum_{i=1}^{\tau} \ell_k^i(w_k^i) - \min_{w_k} \sum_{t=1}^{\tau} \ell_k^t(w_k) \leq U_k(\tau)$

- Running loss $L_k(\tau) := \frac{1}{\tau} \sum_{i=1}^{\tau} \ell_k^i(w_k^i)$,
- Using A2: $L_k(\tau) - \frac{U_k(\tau)}{\tau} \leq \min_{w_k} \frac{1}{\tau} \sum_{i=1}^{\tau} \ell_k^i(w_k)$.

Empirical loss minimizer

M-LCB algorithm

Confidence bounds

- $L_k(\tau) - \frac{U_k(\tau)}{\tau} \leq \min_{w_k} \frac{1}{\tau} \sum_{i=1}^{\tau} \ell_k^i(w_k),$
- Connect **empirical loss** minimizer with $\min_{w_k} \mathbb{E}_{\xi} \ell(g_k(w, \xi), \xi).$

Confidence Bounds for **minimizer**

$$\text{LCB}(k, \tau, \delta) := L_k(\tau) - \frac{U_k(\tau)}{\tau} - c_1 \sqrt{\frac{\log(t/\delta)}{t}},$$

$$\text{UCB}(k, \tau, \delta) := L_k(\tau) + c_2 \sqrt{\frac{t/\delta}{t}}.$$

M-LCB algorithm

Confidence bounds

Price of learning

$$\text{LCB}(k, \tau, \delta) := L_k(\tau) - \frac{U_k(\tau)}{\tau} - c_1 \sqrt{\frac{\log(\tau/\delta)}{t}},$$

$$\text{UCB}(k, \tau, \delta) := L_k(\tau) + c_2 \sqrt{\frac{\tau/\delta}{t}}.$$

Price of uncertainty

Lemma. Let $\delta \in [0,1]$ be given. Then with probability $\geq 1 - \delta$

for all $\tau > 0; k \in [K]$:

$$\text{LCB}(k, \tau, \delta) \leq \min_{w_k} \mathbb{E}_{\xi} \ell(g_k(w, \xi), \xi) \leq \text{UCB}(k, \tau, \delta).$$

M-LCB algorithm

- For $t = \overline{1, \dots, T}$:

- Choose arms set $S_t = \arg \min_{S \subset [K], |S|=M} \sum_{k \in S} \text{LCB}(k, \tau_k^t, \delta)$, <Top M arms with lowest LCB

- $a_t = \arg \min_{k \in S_t} \text{UCB}(k, \tau_k^t, \delta)$, <The most confident arm in S_t

- Receive data ξ_t , suffer loss $\ell_{a_t}^t(w_{a_t}^t)$,

- Observe losses for $a \in S_t$. <Arms parameters update

M-LCB algorithm

- For $t = \overline{1, \dots, T}$:

- Choose arms set $S_t = \arg \min_{S \subset [K], |S|=M} \sum_{k \in S} \text{LCB}(k, \tau_k^t, \delta)$, <Top M arms with lowest LCB

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- Receive data ξ_t , suffer loss $\ell_{a_t}^t(w_{a_t}^t)$,

- Observe losses for $a \in S_t$.

<Arms parameters update

Actually, here is a 'smoothing' trick
in algorithm should be used

M-LCB algorithm

Aim

$$R(T) := \sum_{t=1}^T \ell_{a_t}^t(w_{a_t}^t) - T \min_{k \in [K]} \min_{w_k} \mathbb{E}_{\xi} \ell(g_k(w_{\downarrow} \xi), \xi) \rightarrow \min,$$

Best fitted model Expected loss

↓

Regret bounds

Theorem. Let $\alpha, \delta \in (0, 1)$. Suppose each expert k satisfies A2

with $U_k(t, \delta) = O(t^\alpha c(\delta))$. Then, with probability at least $1 - \delta$ the regret of M-LCB is bounded for all $T \geq 1$ as

$$R(T) \leq O \left(\sqrt{\frac{KT}{M} \log\left(\frac{KT}{\delta}\right)} + \left(\frac{K}{M}\right)^{1-\alpha} T^\alpha c(\delta) \right).$$

Uncertainty regret

Model convergence regret

Comparison

- Learnable — is algorithm consider learning arms
- Multi-arm — is algorithm consider multiple updates per step
- Multiple play — is algorithm updates mean top reward (robustness)

Algorithm / Reference	Learnable	Multi-arm	Multiple-play	Regret rate (up to logs)
CORRAL + smoothing wrapper [24]	✓	×	×	$\tilde{O}\left(\sqrt{KT} + K^\alpha T^{1-\alpha} + K^{1-\alpha} T^\alpha c(\delta)\right)$
EXP3.P + smoothing wrapper [24]	✓	×	×	$\tilde{O}\left(\sqrt{KT} + K^{\frac{1-\alpha}{2-\alpha}} T^{\frac{1}{2-\alpha}} c(\delta)^{\frac{1}{2-\alpha}}\right)$
Dynamic Balancing [9]	✓	×	×	$\tilde{O}\left(\sqrt{KT} + K^{1-\alpha} T^\alpha c(\delta)\right)$
Prediction with Limited Advice [25]	×	✓	×	$\tilde{O}\left(\sqrt{\frac{KT \log K}{M}}\right)$
H-INF [33]	×	✓	×	$\tilde{O}\left(\max\{\sqrt{KT/M}, \sqrt{T \log K}\}\right)$
UCB-based multiple-play bandits [20]	×	×	✓	$\tilde{O}\left(\sqrt{KT/M}\right) \alpha = \frac{1}{2}$
M-LCB [this work]	✓	✓	✓	$\tilde{O}\left(\sqrt{\frac{KT}{M}} + (K/M)^{1-\alpha} T^\alpha c(\delta)\right)$

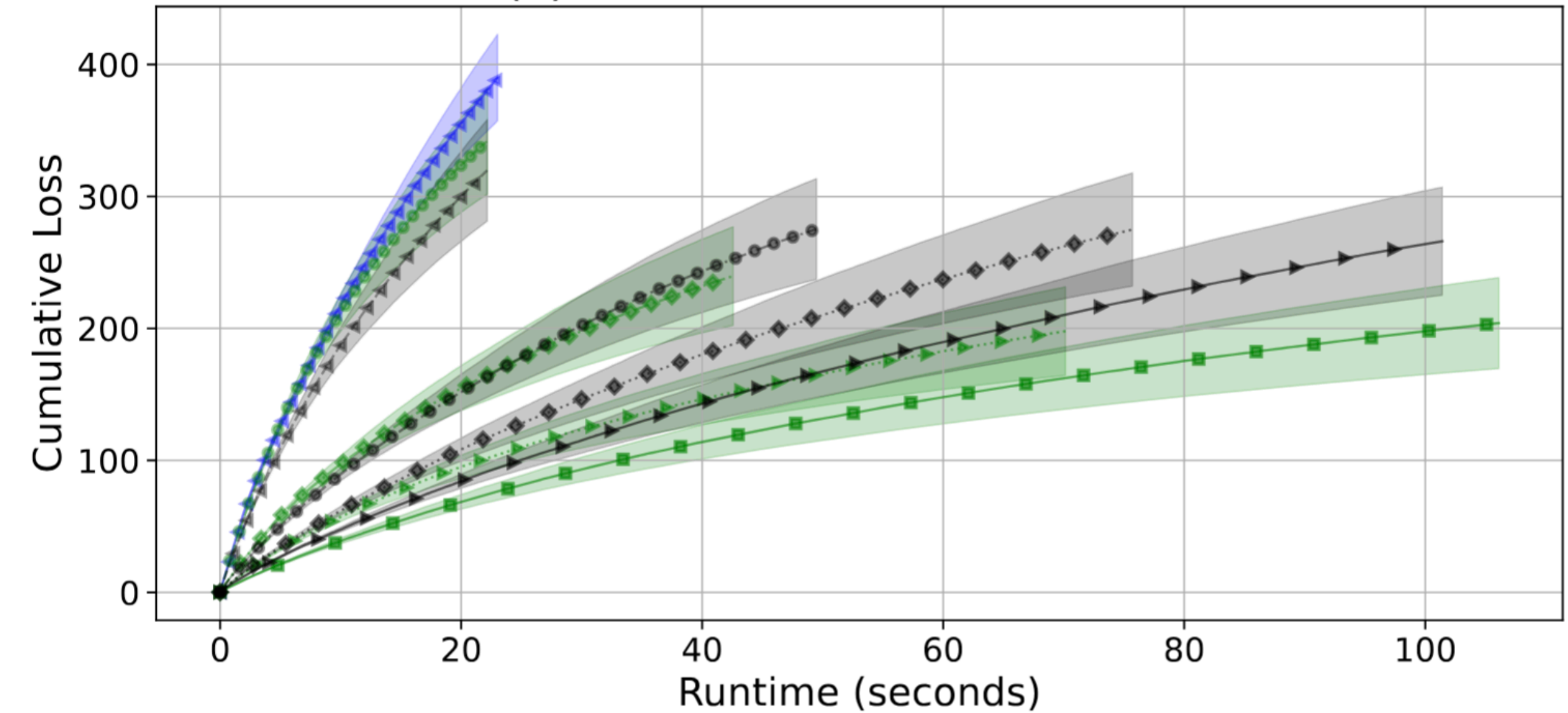
Experiment

As arm consider different functions $\{f_i\}_{i=1}^{10}$

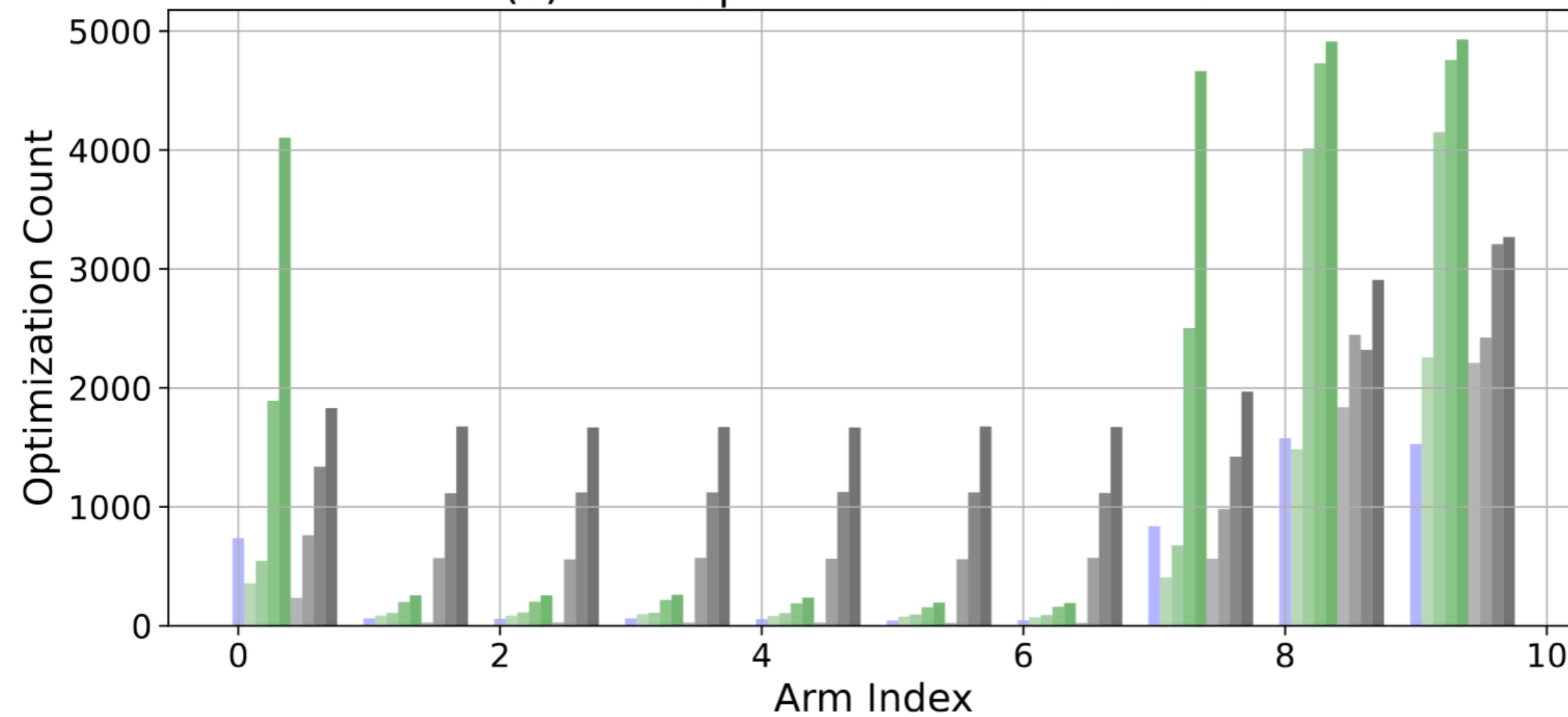
Generate data by $y_i = f_9(x_i^T w)$

f_7, f_8 are the most similar to it

(a) Cumulative Loss vs Runtime



(c) Arm Optimization Distribution



(b) Arm Selection Distribution



Выносятся на защиту

1. Протокол задачи обучения набора моделей в форме MAV
2. Алгоритм M-LCB
3. Теоретические гарантии на работу алгоритма

На данный момент статья подана в A^* конференцию, проходит рецензирование